Class 14: Active Portfolio Management I

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Paul Goldsmith-Pinkham
Yale University

https://paulgp.com

Yale School of Management

Today's Question

A stock-picker claims a Sharpe ratio of 1.5. What do you need to believe for that to be worth paying 2-and-20?

Positioning

Where We've Been	Where We Are	Where We're Going
Factor models; passive vs. active	How to think about active management	Evaluating track records

Learning Objectives

By the end of today's class, you should be able to:

1. Decompose active return into systematic (factor) and idiosyncratic (α) components
2. Explain the information ratio and its role in manager evaluation
3. Describe the Treynor-Black model intuition (without heavy math)

Welcome Back!

Quick recap of the first half of the semester:

- **Markets & Institutions**: How markets work, trading costs
- **Risk & Return**: Measuring performance, Sharpe ratios
- **Portfolio Construction**: Diversification, efficient frontier, tangency portfolio
- **CAPM & APT**: Systematic risk, factor models
- **Evidence**: Anomalies, behavioral biases, limits to arbitrage

Today: What do you do if you think you have an **edge**?

Preview of Our Goal

The CAPM suggests that all investors should hold the market index.

- This view is based on the assumption that all investors share the same information

What if you have extra information?

- How do we combine new information with an otherwise efficient portfolio?
- (What makes the portfolio "efficient"?)

In theory, the Markowitz model will do this, but not for the universe of *investable* assets.

Preview of Our Goal

Instead, we replace full-blown portfolio optimization (Markowitz) with a single-factor model based on the CAPM.

- Using inputs from the model, we can build an active portfolio
  - This is called the “satellite portfolio”
- Optimally combine this active portfolio with the market index
  - This is called the “core portfolio”

This two-part procedure is the **Treynor-Black model**: flexibility for alpha/beta views while remaining diversified.

Computation Costs of Full Portfolio Optimization

Suppose we wish to perform full Markowitz optimization for the S&P 500.

How many parameters would we need?

Parameter	General Number	S&P
σ_i^2	N	500
$E(r_i)$	N	500
$\sigma_{i,j}$	$\frac{1}{2} \times N \times (N-1)$	124,750
Total	$\frac{1}{2} \times N \times (N+3)$	**125,750**

What about the Wilshire 5000 (all stocks)?

A Single Factor Alternative

We can use a CAPM/single-factor relation to simplify inputs.

The CAPM regression (BKM call this the "Single Index Model"):

$$r_{i,t} - r_f = \alpha_i + \beta_i \left(r_{m,t} - r_f \right) + \varepsilon_{i,t}$$

$$\varepsilon_{i,t} \sim \mathcal{N}\left(0, \sigma_i^2(\varepsilon)\right)$$

- Idiosyncratic variance $\sigma_i^2(\varepsilon)$ varies across stocks
- We assume $\mathrm{Cov}\left(\varepsilon_{i,t}, \varepsilon_{j,t}\right) = \mathrm{Cov}\left(\varepsilon_{i,t}, r_{m,t}\right) = 0$

What does this buy us?

Single Index Model: Expected Returns

Taking expectations of both sides gives the SML:

$$E(r_i - r_f) = \alpha_i + \beta_i E(r_m - r_f)$$

Use betas to get variances and covariances:

$$\sigma_{i,j} = \mathrm{Cov}(r_i, r_j) = \beta_i \beta_j \sigma_m^2, \quad i \neq j$$

$$\sigma_i^2 = \beta_i^2 \sigma_m^2 + \sigma_i^2(\varepsilon)$$

Why is this helpful?

Single Index Model: Parameter Count

Now we only need a handful of parameters!

Parameter	General Number	S&P
$E(r_m)$	1	1
r_f	1	1
σ_m^2	1	1
α_i	N	500
β_i	N	500
$\sigma_i^2(\varepsilon)$	N	500
Total	$(N+3)$	**1,503**

A lot easier than 125K parameters!

Single Index Model: Parameter Count

Pop Quiz: If alphas are all zero, and all assets are included, what portfolio do we choose?

Modifying Inputs with a Single-Index Model

Single index model brings both **parsimony** and **discipline** to inputs.

CAPM presents an equilibrium view of expected returns vs. reliance on messy historical returns.

Consider the following exercise:

1. Estimate expected returns and covariances from the data
2. Calculate the optimal market portfolio
3. Construct each stock's beta with the market portfolio
4. Back out the expected returns from the model and compare

What do we miss by doing this?

We can take the input from the CAPM — plus or minus alpha...

- What if we need to adjust betas as well?

Treynor-Black Model

The single-index model gives us the machinery. Now: how do we *use* it when we believe we have alpha on a few stocks?

So far, we have looked at a small number of assets for convenience.

In reality, even the most diligent analysts will only have valuable views on a **handful** of assets.

How do we achieve benefits of diversification while still focusing our views on a handful of stocks?

Treynor-Black Model: Core/Satellite

Treynor-Black suggests a two-stage, **core/satellite** approach:

1. **First stage:** Build active portfolio based on alpha estimates
2. **Second stage:** Decide how much to allocate to active portfolio vs. the index

The portfolio will **maximize the Sharpe ratio** of the total portfolio.

Treynor-Black: Active Portfolio Weights

Construct the active portfolio (A) using alpha estimates and observed residual variance:

$$w_i^0 = \frac{\alpha_i / \sigma_i^2(\varepsilon)}{\sum_j \alpha_j / \sigma_j^2(\varepsilon)}$$

- w_i^0: optimal weight on stock i within the active portfolio A
- α_i: expected excess return over the CAPM prediction for stock i
- $\sigma_i^2(\varepsilon)$: residual (idiosyncratic) variance of stock i

How do we interpret this equation?

- Overweight stocks with high alpha *relative to* their idiosyncratic risk

How Much to Allocate to the Active Portfolio?

Next: how much to put in the active portfolio, w_A?

$$w_A = \frac{\alpha_A/\sigma_A^2(\varepsilon)}{E(r_m - r_f)/\sigma_m^2 + (1-\beta_A)(\alpha_A/\sigma_A^2(\varepsilon))}$$

where

$$\beta_A = \sum_{i\in A} w_i\beta_i, \quad \alpha_A = \sum_{i\in A} w_i\alpha_i, \quad \sigma_A^2(\varepsilon) = \sum_{i\in A} w_i^2\sigma_i^2(\varepsilon)$$

- What happens as α_A gets large? Or σ_A^2 gets small?
- What about the market's Sharpe ratio?

Final active portfolio weight on stock i: $w_i^A \times w_A$

$(1 - w_A)$ is invested in the passive index.

Example

Market returns are 9% with $\sigma_m^2 = 4\%$ and $r_f = 3\%$.

Combine the index with the following beliefs about Nvidia and Shopify:

	α	β	$\sigma_i^2(\varepsilon)$
NVDA	0.5%	2	7%
SHOP	0.2%	1.75	12%

What are we assuming about the other stocks?

$$w^0_{\text{NVDA}} = \frac{\alpha_{\text{NVDA}}/\sigma^2_{\text{NVDA}}(\varepsilon)}{\alpha_{\text{NVDA}}/\sigma^2_{\text{NVDA}}(\varepsilon) + \alpha_{\text{SHOP}}/\sigma^2_{\text{SHOP}}(\varepsilon)} = 0.81, \quad w^0_{\text{SHOP}} = 0.19$$

Example (Continued)

Based on these weights:

$$\beta_A = 0.81 \times 2 + 0.19 \times 1.75 = 1.95$$
$$\alpha_A = 0.81 \times 0.005 + 0.19 \times 0.002 = 0.0044$$
$$\sigma_A^2(\varepsilon) = 0.81^2 \times 0.07 + 0.19^2 \times 0.12 = 0.05$$

This gives us the active weight:

$$w_A = \frac{0.0044/0.05}{0.06/0.04 + (1 - 1.95)(0.0044/0.05)} = 0.06$$

Example: Final Allocation

So we invest roughly **6%** in an active portfolio, and **94%** in the index.

Given the active portfolio weights in NVDA and SHOP, we get final portfolio weights:

- **94%** for the index
- **5%** for NVDA
- **1%** for SHOP

Key takeaway: Even with strong alpha views, the optimal allocation to active positions is *modest*.

The Information Ratio

The **information ratio** (IR) measures the quality of active management:

$$\mathrm{IR} = \frac{\alpha_p}{\sigma(\varepsilon_p)}$$

- Numerator: the alpha (excess return over benchmark)
- Denominator: the tracking error (volatility of active returns)

The IR tells you how much active return you get **per unit of active risk**.

Connection to Treynor-Black: the optimal portfolio maximizes the **squared** Sharpe ratio, which equals:

$$S_P^2 = S_M^2 + \mathrm{IR}^2$$

Back to Today's Question

A stock-picker claims a Sharpe ratio of 1.5. What do you need to believe for that to be worth paying 2-and-20?

- What is the manager's *alpha* after fees?
- What is the *information ratio* after accounting for the 2% management fee and 20% performance fee?
- How much of your portfolio should be allocated to this manager (Treynor-Black)?

Next Class: March 26

Topics: Active Portfolio Management II

- Skill vs. luck vs. cheating
- Active Share
- Insider trading detection from returns

Matt Levine: “Active Management and Insider Trading”